

LLM Week 7 GA

1) Consider a vocabulary $\mathcal{V}$,

$\mathcal{V} = \{\text{[start]}, \text{building}, \text{character}, \text{a}, \text{is}, \text{astronomy}, \text{science}, \text{experience}, \text{natural}, \text{[end]}\}$.

Assume that the BART model is being trained using BART's denoising objective. Suppose the original sentence is "[start] astronomy is a character building experience [end]" and the corrupted sentence is "[start] astronomy [mask] [mask] character building [mask] [end]"

The prediction probabilities outputted by the model are given below.

$$\hat{Y} = \begin{bmatrix} \text{s} & \text{b} & \text{c} & \text{a} & \text{i} & \text{s} & \text{e} & \text{n} & \text{e} \\ 0.05 & 0.02 & 0.14 & 0.07 & 0.09 & 0.41 & 0.08 & 0.05 & 0.08 & 0.01 \\ 0.10 & 0.04 & 0.06 & 0.16 & 0.43 & 0.01 & 0.06 & 0.09 & 0.01 & 0.04 \\ 0.05 & 0.07 & 0.28 & 0.29 & 0.03 & 0.08 & 0.04 & 0.08 & 0.04 & 0.04 \\ 0.10 & 0.02 & 0.48 & 0.02 & 0.01 & 0.07 & 0.06 & 0.14 & 0.07 & 0.01 \\ 0.08 & 0.29 & 0.04 & 0.01 & 0.04 & 0.03 & 0.25 & 0.05 & 0.15 & 0.06 \\ 0.14 & 0.22 & 0.13 & 0.08 & 0.01 & 0.03 & 0.06 & 0.23 & 0.08 & 0.03 \\ 0.1 & 0.01 & 0.01 & 0.01 & 0.07 & 0.2 & 0 & 0.25 & 0.05 & 0.3 \end{bmatrix}$$

$$\begin{array}{lcl} 0.41 & \rightarrow & -0.89 \\ 0.43 & \rightarrow & -0.84 \\ 0.29 & \rightarrow & -1.23 \\ 0.48 & \rightarrow & -0.733 \\ 0.29 & \rightarrow & -1.23 \\ 0.23 & \rightarrow & -1.46 \\ 0.3 & \rightarrow & -1.2 \end{array} \left. \vphantom{\begin{array}{lcl} 0.41 \\ 0.43 \\ 0.29 \\ 0.48 \\ 0.29 \\ 0.23 \\ 0.3 \end{array}} \right\} \frac{\text{sum}}{7}$$

The zeroth row of the matrix is the output probability distribution by the model given the special [Start] token as input. Following the Greedy Search decoding strategy, the subsequent rows give the conditional probability distribution conditioned over the previous tokens. What is the loss value?

0.9

2) Assume that we have 12 sufficiently large supervised NLP datasets for different tasks like sentiment classification, textual entailment, language understanding and so on. Suppose we take all the samples (dropping labels) from these datasets to train the GPT model using the CLM objective. Then which of the following transfer learning approaches is (are) appropriate to transfer the knowledge to any of these 12 downstream tasks? 2 points

- ☒ Zero shot learning
- ☒ Few-shot learning
- ☒ Full Fine-tuning
- ☐ None of the given approaches are appropriate as the model has already been trained using samples from all the 12 datasets

Yes, the answer is correct.

Score: 2

Accepted Answers:

Zero shot learning
Few-shot learning
Full Fine-tuning

3) The strikethrough words in the passage given below denote the words to be dropped from the original sentence. 2 points

"At the heart ~~(of science is)~~ an essential balance between two seemingly ~~contradictory attitudes~~ — an openness to new ideas, no matter how bizarre or ~~(counterintuitive they may be,~~ and the most ruthless skeptical scrutiny of all ideas, old and new. This is ~~(how deep truths are winnowed)~~ from deep nonsense"

Which of the following represents the correct target sequence, with sentinel tokens, to the baseline model that uses the pre-training denoising objective? The characters inside the square brackets are the sentinel tokens and [z] represents the end of the sentinel token in a sentence

- ☒ [v] of science is [w] contradictory attitudes [x] counterintuitive they may be [y] how deep truths are winnowed [z]
- ☒ [a] of science is [b] contradictory attitudes [c] counterintuitive they may be [d] how deep truths are winnowed [z]
- ☐ [a] of science is [b] contradictory attitudes [c] counterintuitive they may be [d] how deep truths are winnowed [e]
- ☐ [v] of science is [w] contradictory attitudes [x] counterintuitive they may be [y] how deep truths are winnowed

at end
there must be
a z
turn alphabetic

4) The statement that "The special sentinel tokens used in the denoising objective of the T5 model are part of the vocabulary" is

- ☒ True
- ☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

5) A dataset contains 66 Billion tokens. Suppose we want to train a model for 1 million steps so that the model will see at least 5 Billion tokens. Then which of the following is appropriate $B \times T$ 1 point

- ☐ 8×128 1024 x
- ☒ 64×128 8192 ✓ 66 x 10⁹
- ☒ 128×512 65536 ✓
- ☒ 4×2048 8192 ✓
- $1 \times 10^6 \rightarrow 5 \times 10^9$
- ≥ 5000

6) Consider fine-tuning the T5 model on a sentiment classification task. The T5 model was pre-trained using the C4 dataset. Suppose, in the original dataset, the positive samples are denoted by integer 1 and the negative samples are denoted by integer 0. Choose the necessary modifications that one needs to perform to use the T5 model for fine-tuning **2 points**

- ☒ Map the label 0 to "negative" and 1 to "positive" in the dataset
- ☒ Map the label 0 to "0" and 1 to "1" in the dataset
- ☐ Update the vocabulary of T5 with the modified labels
- ☐ Add at least one linear layer on top of the decoder for final classification

convert it
to a string

Yes, the answer is correct.

Score: 2

Accepted Answers:

Map the label 0 to "negative" and 1 to "positive" in the dataset

Map the label 0 to "0" and 1 to "1" in the dataset

7) Suppose we have two models. A decoder model D with P parameters and an encoder-decoder model E with $2P$ parameters. Then both models' computational costs (under the T5 framework) are the same. The **1 point** statement is

☒ True

☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

$D \rightarrow P \rightarrow M$
 $E \rightarrow 2P \rightarrow M$